

Telecom Churn Prediction

Predicting high-risk customers and automating retention strategies to maximise profit.

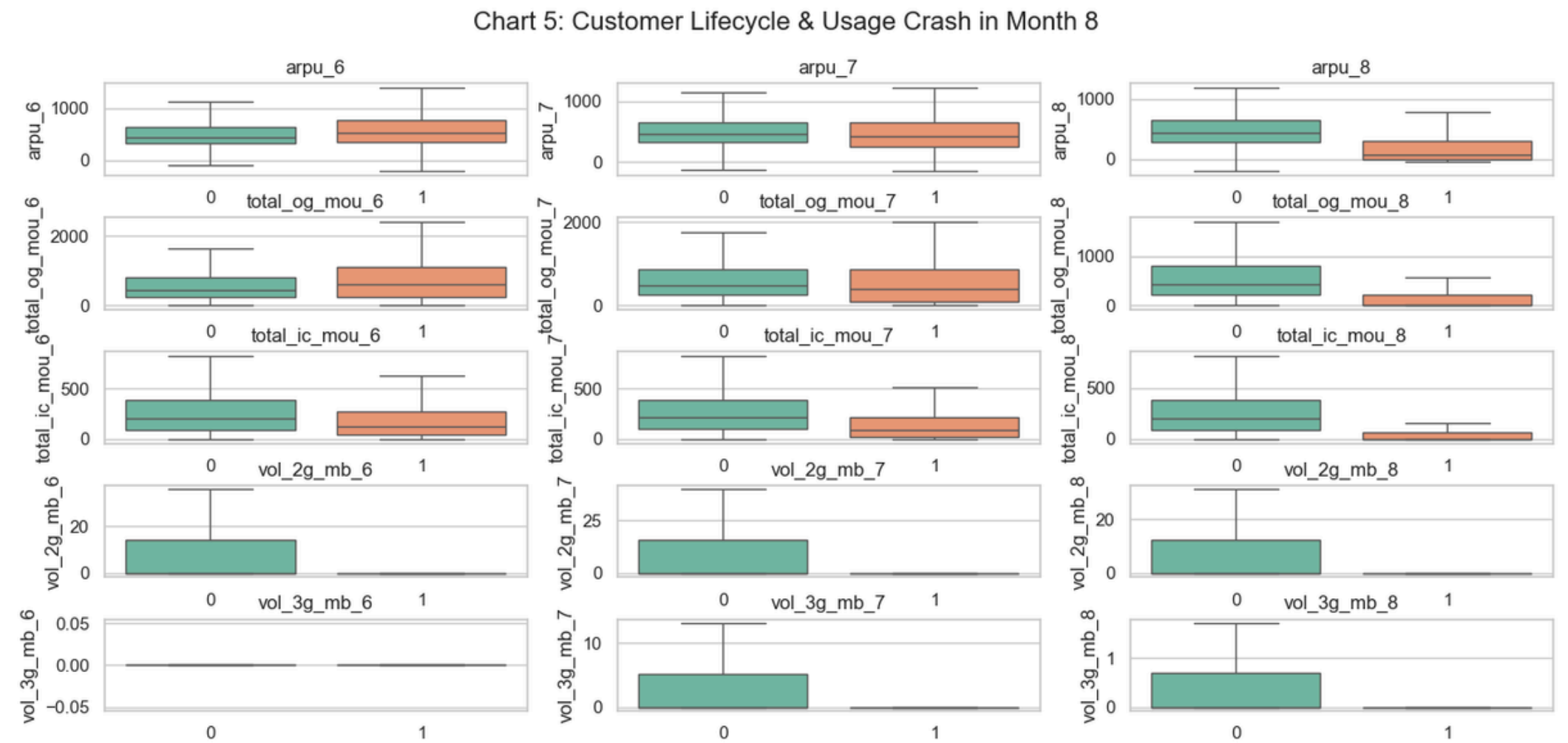
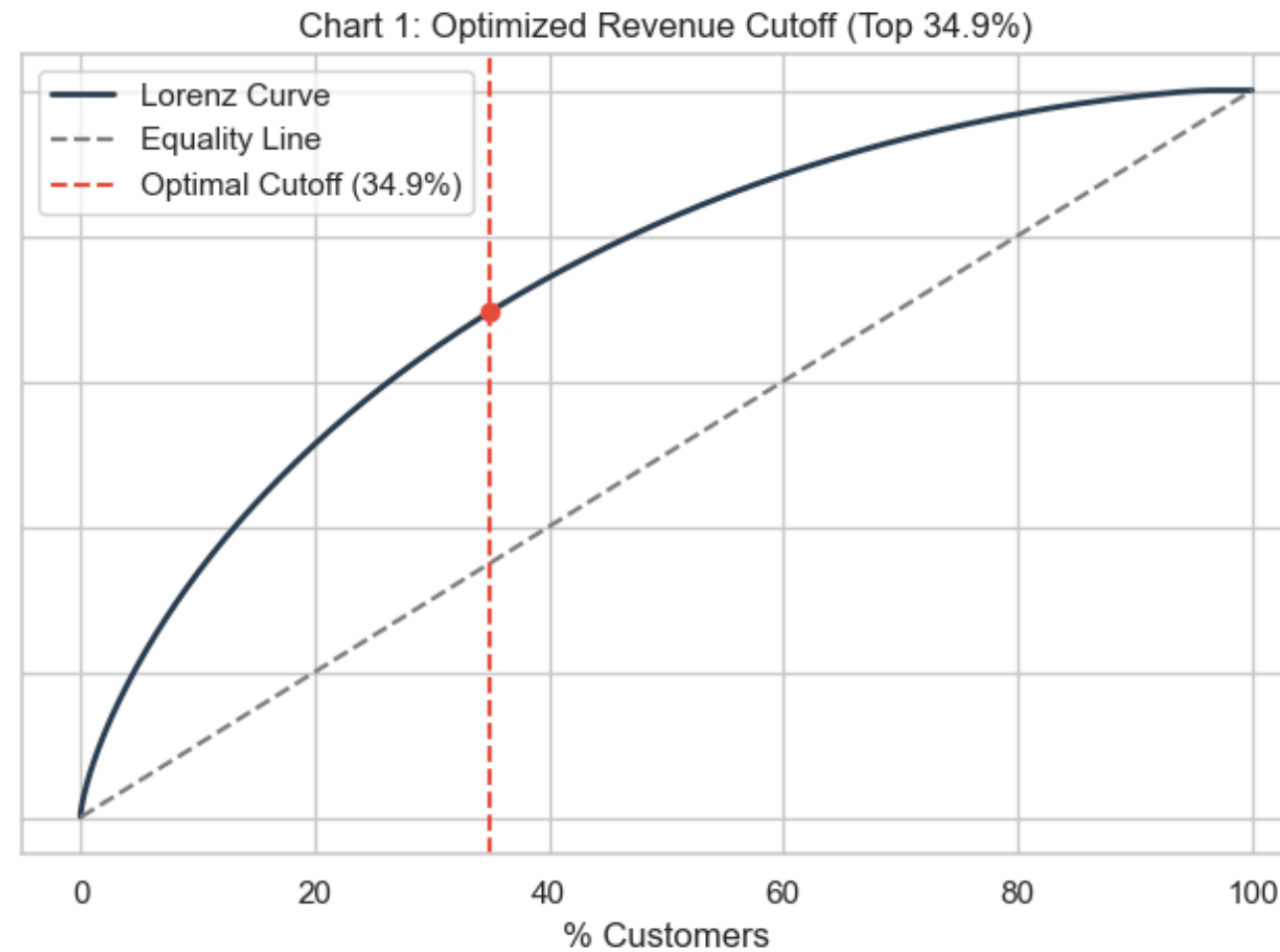
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About the Dataset

- Kaggle dataset, sourced from a recruitment hackathon
- Over 1 lakh customer records with 150+ variables
- Spread across 4 months, from June to September
 - Month 6 and 7 are the good phase, the baseline for usage
 - Month 8 is the action phase, customer thinks about churning
 - Month 9 is the churn phase, the customer has churned
- 8% churners – Defined as zero usage

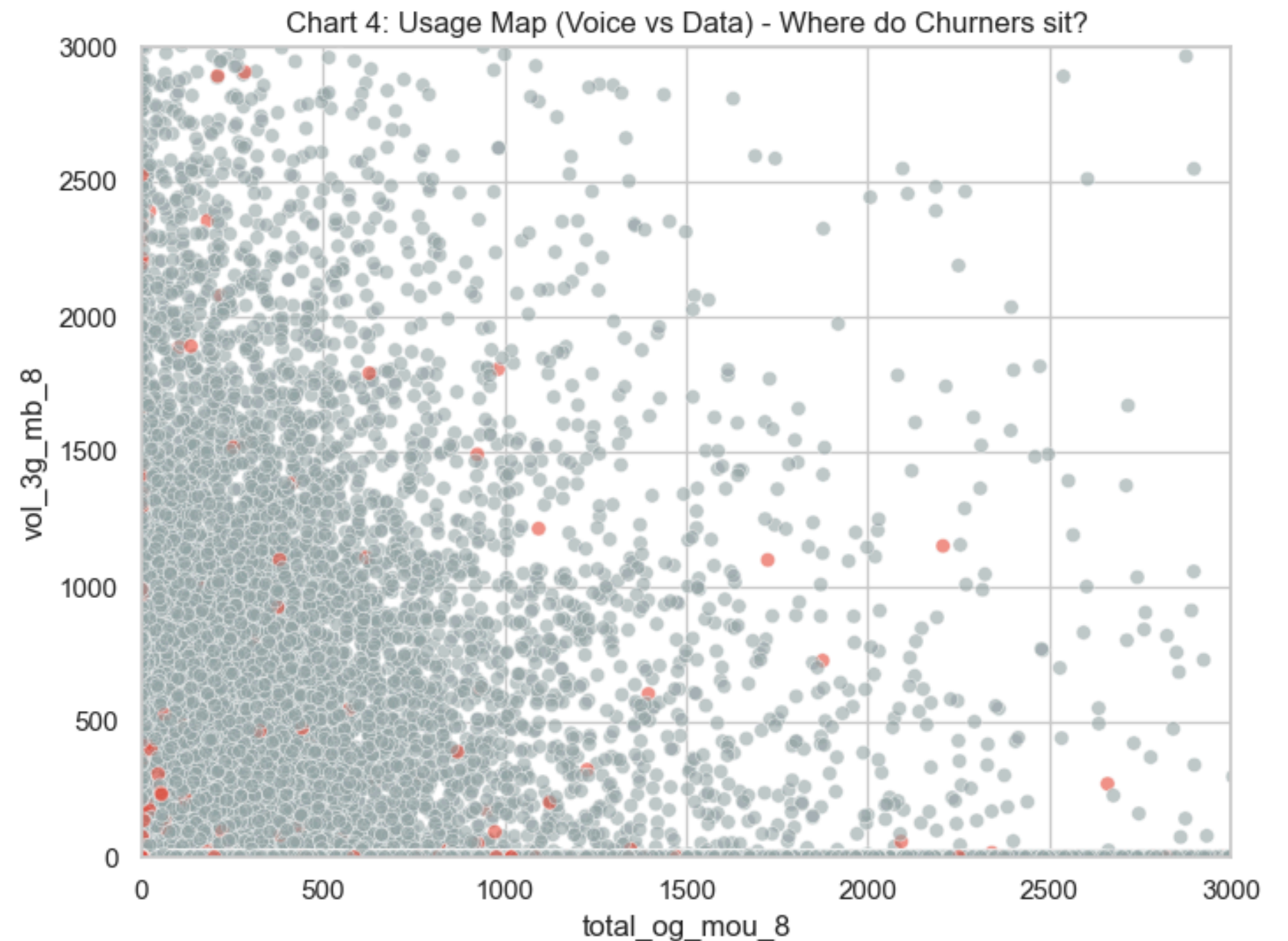


Data Pre-Processing

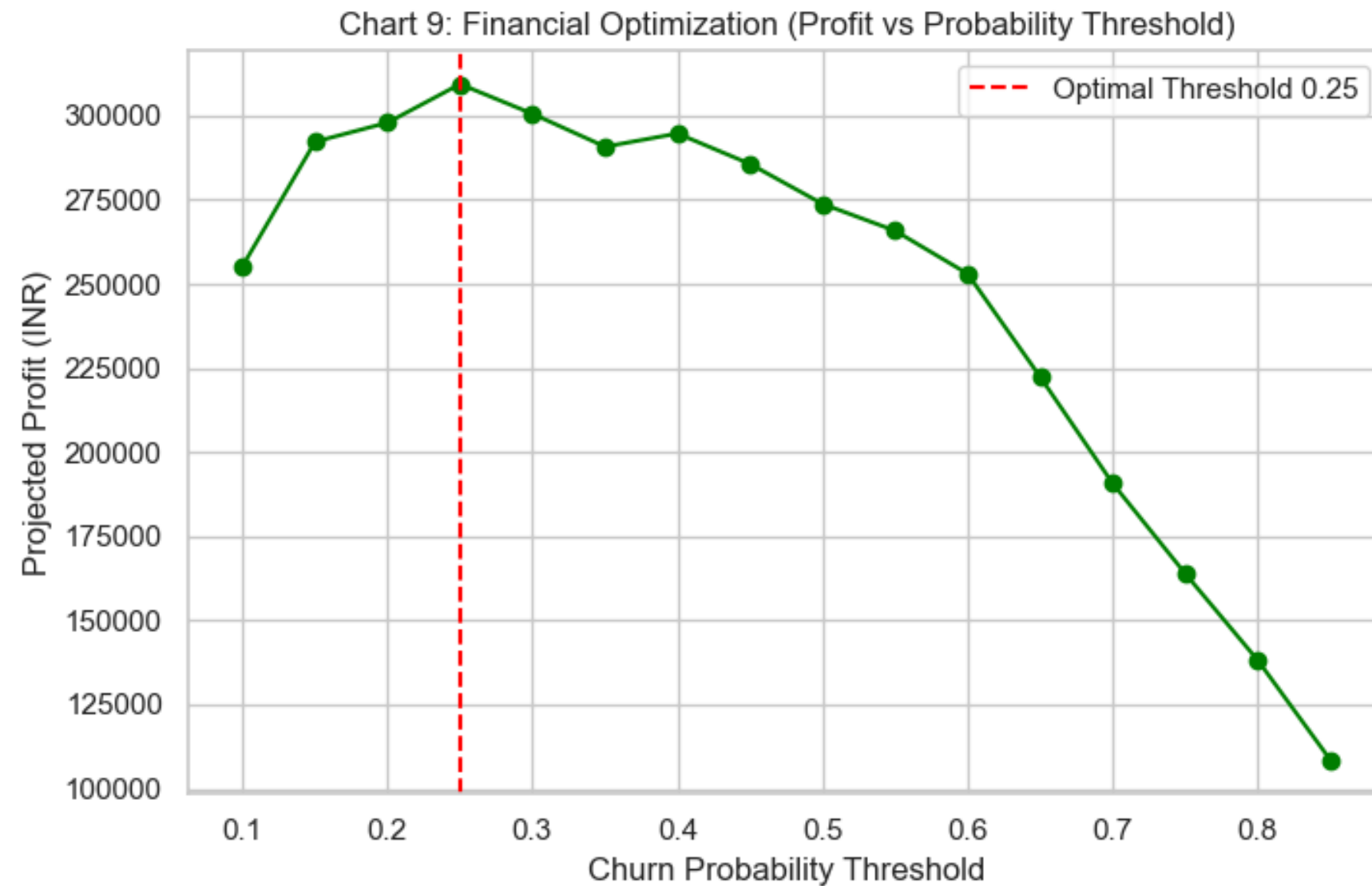
- 70% null values: logically filled with zeros, rest dropped
- Feature Engineering: Rate of change of usage was calculated
- High-Value Customers: Top 34.9% from Lorenz Curve Analysis
- High Multicollinearity: Variance Factor Inflator test and Correlation analysis
- Principal Component Analysis: From 150+ variables to 50, captured 95% variance

Model Building

- Augmented Dickey-Fuller test was conducted to check for stationarity, confirmed stable, reliable patterns
- Logistic Regression was trained with balanced class_weight parameter
- Random Forest Classifier was used on uncompressed variables to extract Feature Importance
- K-Means clustering, with the help of the Elbow Method, of churners for targeted retention strategies



Financial Simulator



Inputs:

- Cost of Retention Offer (10% ARPU)
- Estimated Success Rate of Offer (50%)
- Lifetime Value Saved (6 months)

Output:

- Net Profit across all thresholds
- Reveals most profitable threshold/probability

Results and Interpretation



ROC – AUC = 0.90



PR – AUC = 0.51; Baseline = 0.08



Optimal Threshold = 0.25



Precision = 0.17; Recall = 0.95



Connectivity and Community (Incoming) are the strongest predictors; Word of Mouth most important due to similar services

Chart 8: Precision-Recall Curve (Imbalance Robust)

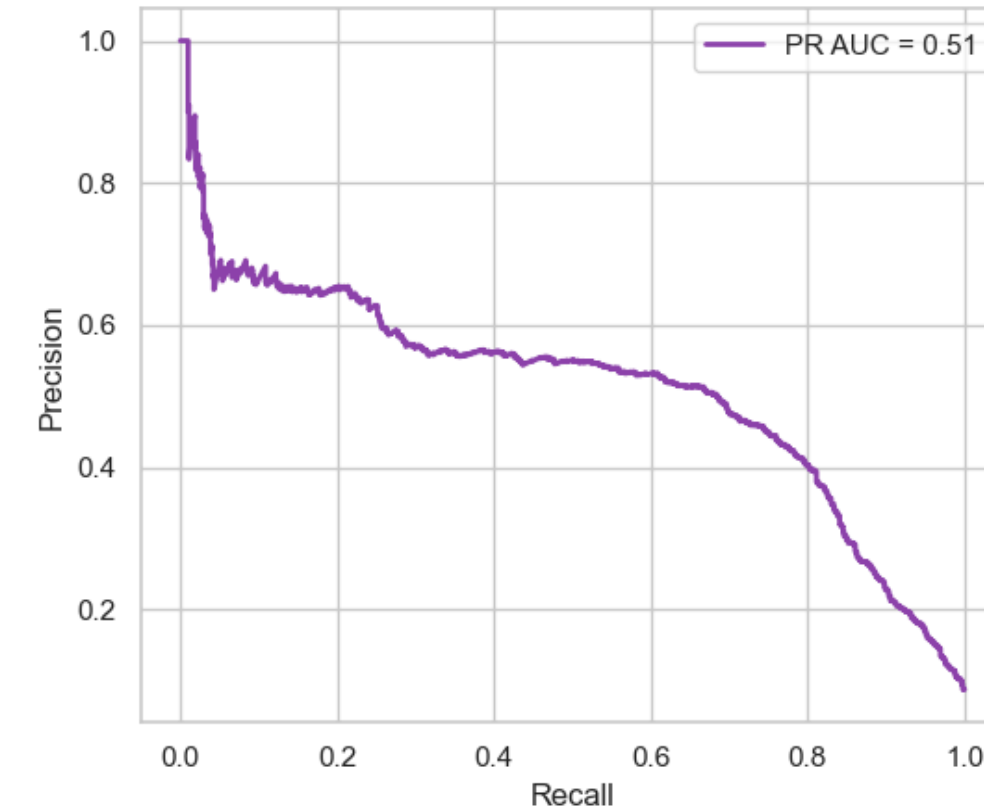
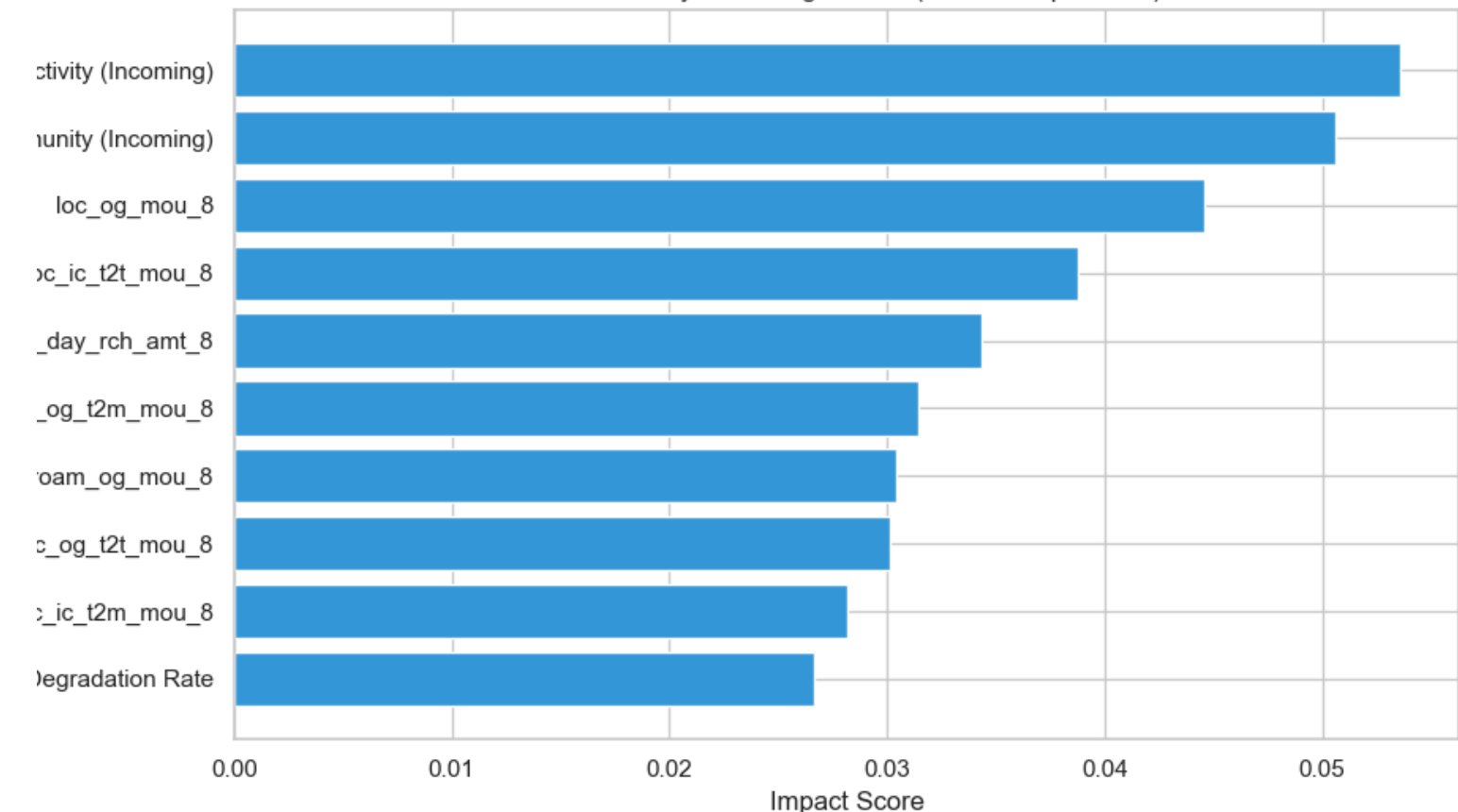
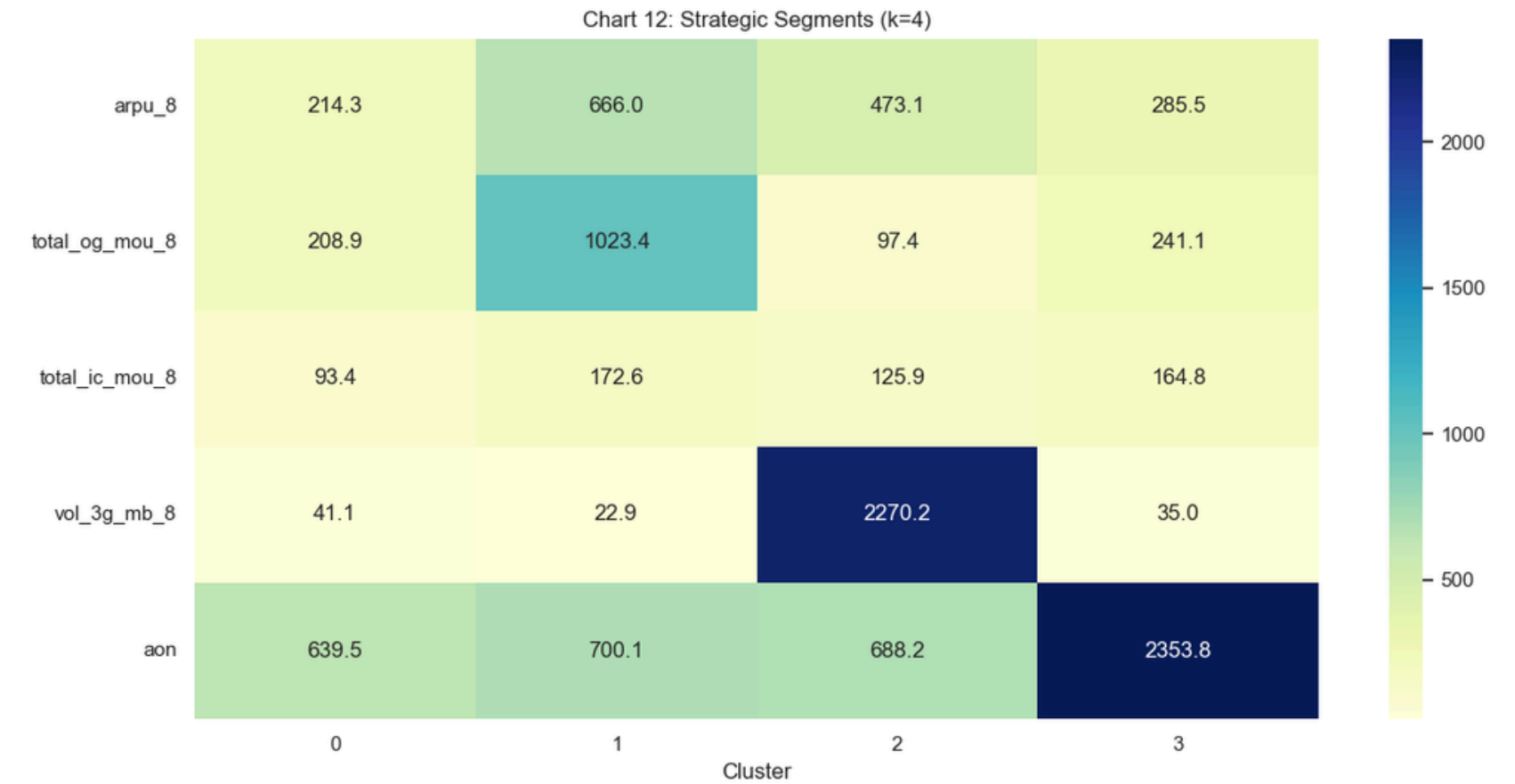


Chart 10: Key Marketing Drivers (Feature Importance)



Retaining Churners

CHURNER PERSONAS AND RECOMMENDED STRATEGIES



01

LOYAL CUSTOMERS

Loyalty rewards like dedicated helpline, free upgrades, free subscriptions, etc

02

VOICE USERS

Discounted or free voice packs

03

DATA USERS

Free Booster packs or prioritised speeds

04

PRICE SENSITIVE USERS

Discounted validity extensions

Conclusion

01

Future Scope

Natural Language Processing of customer support

03

Business Impact

Proactively recovers losing revenue while preventing unnecessary expenditure

02

Ethical and Policy Impact

Should obtain clear consent according to DPDP Act; No manipulation or surveillance

04

Societal Impact

Churn indicates underlying problems, helps maintain stable and reliable infra

Thank You!

